a new term was coined for it in 2020—MAFLD, or metabolic dysfunction-associated fatty liver disease. It signifies not just fat accumulation in the liver but gestures at the whole cluster of conditions contingent upon each other, like obesity, insulin resistance or diabetes. A worsening of liver, then, can cause or worsen metabolic diseases, and also be affected by them.

And it's not just the metabolic system that suffers on account of an ailing liver. "The liver purifies your blood. If it isn't working properly, it can impact a variety of organs, from your heart to your brain to your gut," says Dr Sarin. A 2023 study in the journal *Frontiers in Endocrinology* found loss of brain function when the liver is unable to remove blood toxins. Another 2019 study in the *European Journal of Clinical Investigation* found a link between fatty liver and gastroesophageal reflux disease, which causes heartburn, acid reflux and regurgitation. It also noted that a fatty liver leads to poor health of the gut microbiome.

HOW TO PREVENT AND TREAT FATTY LIVER

Contrary to popular perception, a low-fat diet is not enough to prevent NAFLD. You don't need to eat fat for fat to be created in your body; excess calories, regardless of their source, are enough. And while excess saturated fats and transfats are harmful to the liver, recent studies indicate the essential fatty acid C15:0 (found in cow's milk) may actually have a beneficial effect--and that its deficiency may be a contributing factor in NAFLD. But experts are worried about the rise in internet-guided fad diets for health or weight loss, like high-fat ketogenic diets, which, in the absence of professional supervision, can lead to a huge increase in dietary fat without commensurate effort to burn it in the body.

Interlinked metabolic conditions can also predispose a person to NAFLD. Diabetes, for instance, worsens fatty liver, as does high cholesterol. The reverse is also true. "Type 2 diabetes is often accompanied by insulin resistance, which can exacerbate fat accumulation in the liver. Elevated blood glucose levels can lead to increased lipogenesis (fat production) and decreased fatty acid oxidation in the liver," says Dr Deepak Lahoti, senior director, gastroenterology, hepatology and endoscopy, Max Super Speciality Hospital, Patparganj, Delhi. Similarly, excess body fat, particularly visceral fat, can harm the liver thanks to the pro-inflammatory compounds released by adipose cells. PCOS (polycystic ovary syndrome) and sleep apnea are also risk factors for NAFLD. "Women with PCOS often experience insulin resistance, leading to elevated insulin levels, which is a risk factor for fatty liver," says Dr Lahoti. "As for sleep apnea, the intermittent hypoxia may lead to oxidative stress and inflammation, further exacerbating liver damage. Overall, these conditions create a cycle of metabolic dysregulation, leading to increased fat deposition in the liver, inflammation and potential progression to more severe liver diseases."

Prevention, of course, is better than cure but so is diagnosing a health condition early enough. "Regular check-ups and being proactive about liver health can make a significant difference," says Dr Lahoti. But NAFLD symptoms take a long time to show up. Eventually, some people may experience fatigue, a feeling of fullness, nausea, jaundice, dark urine, or mild liver enlargement, by which time they have lived with fat in the liver long enough for it to have an impact on other metabolic and organ functions, particularly diabetes. It is therefore recommended that those with risk factors go for preventive testing. "If you have obesity, diabetes, any other metabolic issue, drink alcohol or a family history of liver disease, then it is important to screen your liver regularly," says Dr Sarin.

Two types of tests can diagnose whether you have NAFLD. The first is an abdominal ultrasound, which is first recommended to avoid needless exposure to radiation and more expensive imaging tests like a CT scan or an MRI. These help visualise the appearance of the liver. Newer tests, such as a transient elastography, are then added for further investigations which include measuring the amount of fat or stiffness in the organ. If advanced liver disease or some other cause for liver issues is suspected, a liver biopsy could be recommended.

Once diagnosed, there is no specific medication for treatment. The usual recommendation is permanent lifestyle

THE LIVER'S BEST DIET

Support your liver with foods that naturally promote health and balance

- **Water:** Drinking water before meals prevents overeating and supports liver function
- **Black Coffee:** Moderate consumption may lower the risk of liver disease
- **Oats & Whole Grains:** Rich in beta-glucans, a soluble dietary fibre, they help reduce obesity and diabetes risks
- **Raw Garlic:** May lower liver cancer risk and reduce fat accumulation
- **Onion:** Helps limit fat storage, potentially preventing fatty liver
- **Dark Leafy Greens:** Provide glutathione, a key antioxidant for liver health
- **Berries:** Dark berries such as blueberries, raspberries and cranberries contain polyphenols that protect against liver damage
- **Plant-based foods:** Low-calorie, plant-based diets support optimal liver function

Compiled from various studies